

Immunological Supply Chain: Self-Healing Supply Chains with AI Digital Antibodies

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Abstract—Modern supply chains are complex, interconnected networks vulnerable to disruptions caused by geopolitical tensions, natural disasters, demand volatility, and supplier failures. Traditional reactive mitigation strategies are insufficient in the face of cascading, multi-domain failures. This paper presents the Immunological Supply Chain (ISC) framework a biologically inspired, end-to-end software prototype that models a supply chain as an adaptive immune system capable of detecting, analyzing, and autonomously responding to disruptions in real time. Drawing a rigorous analogy to biological immunity, the framework maps innate immunity to macro-stress-aware statistical anomaly detection, adaptive immunity to learned Graph Neural Network (GNN) representations and Reinforcement Learning (RL) routing, immunological memory to a FAISS-indexed vector database of 100,000 historical disruptions, and digital antibody agents to autonomous modules for supplier replacement, inventory redistribution, and route recovery. A cytokine storm alert mechanism detects cascade failures from multiple simultaneous anomalies. The system is implemented as a 17-stage pipeline operating on real-world pharmaceutical, logistics, and supply chain datasets totalling over 340,000 records. It constructs a supply chain graph of 2,131 nodes and 23,264 edges, detects 1,535 high-confidence anomalies from 50,000 transactions, achieves a PPO reinforcement learning routing success rate of 100% with reduced average route risk relative to Dijkstra shortest-path, and retrieves relevant historical disruption memory at an average match distance of 0.43. Two interactive web-based dashboards provide visualization and real-time monitoring.

Index Terms—supply chain resilience, anomaly detection, graph neural networks, reinforcement learning, immunological memory, FAISS, digital twins, multi-agent systems, self-healing systems, PPO routing.

I. INTRODUCTION

Global supply chains underpin the delivery of pharmaceuticals, food, electronics, and consumer goods to billions of people. Yet they remain acutely vulnerable to disruptions: the COVID-19 pandemic caused an estimated \$4 trillion in supply chain losses, the 2021 Suez Canal blockage held up \$9.6 billion in trade per day, and semiconductor shortages have persisted for years. In each case, the inability to detect early warning signals, reason about cascading failure, and automatically reconfigure supply paths translated localized incidents into global crises.

Contemporary supply chain management (SCM) systems excel at optimization under normal conditions but lack autonomous adaptability under stress. Rule-based exception management systems trigger alerts after disruptions manifest;

statistical models may detect anomalies but cannot prescribe corrective actions; and human analysts remain the bottleneck for cross-domain decision-making. A fundamentally different architecture is needed one that combines the sensing breadth of statistical methods with the learned pattern recognition of deep learning and the autonomous response capability of multi-agent systems.

This paper proposes and implements the Immunological Supply Chain (ISC) framework, which draws a structural analogy to the vertebrate adaptive immune system. The immune system is the most sophisticated fault-detection and recovery architecture evolved in nature: it operates in real time, maintains memory of past threats, distinguishes self from non-self, deploys targeted responses, and learns from every new encounter. We argue that these same architectural principles layered detection, associative memory, targeted autonomous response, and learning through experience translate directly into the supply chain domain.

A. Biological Motivation

The mammalian immune system operates through two complementary layers. The *innate immune system* provides rapid, non-specific responses: it detects molecular patterns associated with damage or foreign material and raises alarm signals (cytokines) that recruit further defences. The *adaptive immune system* provides antigen-specific responses: B cells and T cells recognise precise molecular signatures, generate clonal armies of effector cells, and retain memory B cells for rapid recall on re-exposure. Clonal selection ensures that only cells matching the antigen proliferate, while outcomes are integrated into immunological memory.

The ISC framework maps each of these mechanisms onto a supply chain function, as summarised in Fig. 1.

B. Contributions

The immune-system analogy for supply chain resilience is not itself new: it has been proposed conceptually [18] and formalised mathematically [19], and a recent systematic review identifies immune-inspired modelling as an emerging but still largely un-operationalised research theme [20]. This paper's contribution is therefore not the analogy, but its operationalisation: to our knowledge, no prior work implements the innate/adaptive immune mapping as a trained, end-to-end

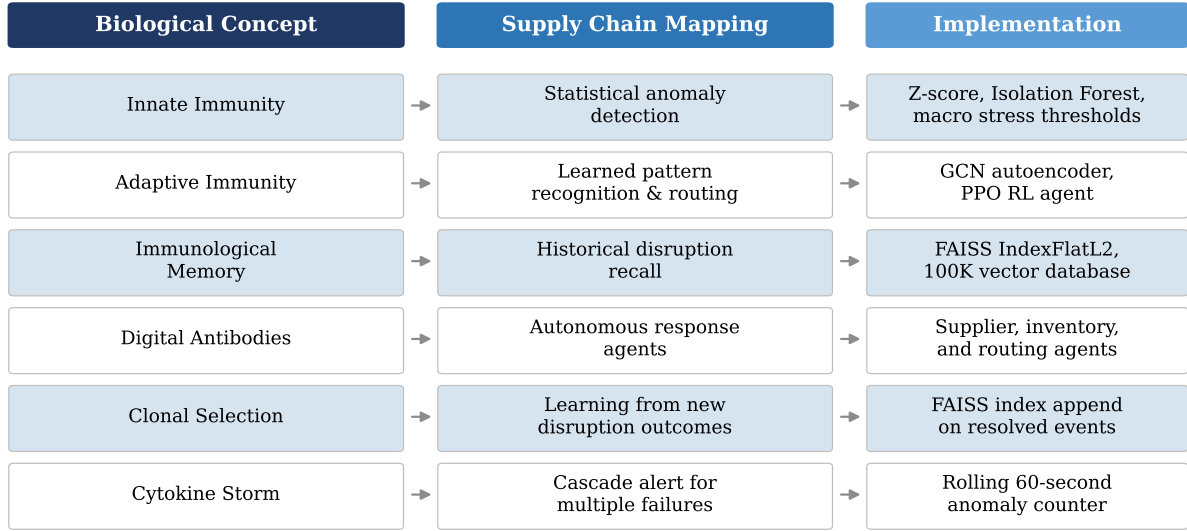


Fig. 1. Biological immune system concepts mapped to supply chain functions and their implementation.

machine learning system validated on real-world data. The primary contributions are:

1. The first (to our knowledge) end-to-end implementation that operationalises the immune-system analogy for supply chains as trained ML components rather than conceptual or purely mathematical constructs: a GCN autoencoder for adaptive recognition, a PPO agent for affinity-maturation-style routing, and a FAISS vector index for immunological memory, integrated in a single pipeline and validated on over 340,000 records across six real datasets.
2. Two specific mechanisms within that pipeline evaluated in isolation: a macro-stress-aware anomaly detection mechanism that dynamically adjusts thresholds based on real-world freight indicator signals, and a custom PPO routing agent with gradient saliency attribution that reduces average route risk relative to shortest-path Dijkstra on 1,115 evaluation scenarios.
3. A more granular operationalisation of the biological analogy than prior conceptual work, additionally implementing clonal selection (online memory update) and cytokine storm (cascade failure alert) as distinct, separately implemented mechanisms.
4. Domain-configurable deployment across pharmaceutical, automotive, and FMCG supply chains, with two interactive web-based dashboards for end-to-end visualization and real-time immune response monitoring.

We report these contributions alongside their limitations (Section IX) rather than as a fully validated system: several components, particularly anomaly detection precision and recovery-time prediction accuracy, show that operationalising the analogy is harder than proposing it, which we view as itself an informative finding.

C. Paper Organisation

Section II reviews related work. Section III details the biological analogy. Section IV describes the datasets. Section V presents the system architecture. Section VI details the methodology with mathematical formulations. Section VII presents experimental results. Section VIII describes the dashboard and real-time system. Section IX discusses strengths, limitations, and future work. Section X concludes.

II. RELATED WORK

A. Supply Chain Resilience and Disruption Management

Supply chain resilience has been extensively studied as an organisational capability [1], [2]. Tang [3] provides a comprehensive review of robust and resilient supply chain strategies, while Hendricks and Singhal [4] demonstrated that supply chain disruptions cause persistent negative stock market effects averaging 40% over a two-year period. More recent work has focused on quantitative disruption modelling [5] and network-theoretic vulnerability analysis [6]. Existing frameworks tend to either optimise supply chain configurations under assumed disruption probabilities, or provide post-hoc analysis tools. The ISC framework differs by operating in real time, learning continuously, and deploying autonomous response agents.

B. Immune-System Analogies for Supply Chain Resilience

The specific analogy this paper builds on mapping innate and adaptive immunity onto supply chain risk response is itself established prior work, not a contribution of this paper. Zheng et al. [18] first proposed a supply chain immune system framework with physical-barrier, innate-response, and adaptive-response layers, illustrated qualitatively on a semiconductor supply chain case study. Ivanov [19] subsequently formalised the same innate/adaptive analogy mathematically, modelling

supply chain resilience as a composition of preparedness and recovery. Neither of these works implements a trained system or validates it against real transactional data. A 2025 PRISMA systematic review of 235 papers on resilient and viable supply chain modelling [20] identifies immune-inspired approaches as an emerging thematic cluster and explicitly notes the absence of methodological frameworks that connect immunological principles to applicable models validated in real operational contexts precisely the gap this paper attempts to address. This paper’s distinction from that prior art is therefore one of operationalisation and empirical validation rather than conceptual origination: we instantiate the same innate/adaptive analogy using a GCN autoencoder, a PPO reinforcement learning agent, and a FAISS vector index, and evaluate the resulting system on six real datasets totalling over 340,000 records, additionally extending the analogy to clonal selection and cytokine storm mechanisms not covered in the layer models above.

C. Anomaly Detection in Supply Chain Data

Statistical anomaly detection using z-scores, moving averages, and control charts is well established in operations management. Isolation Forest [7] has been applied to fraud detection and quality control but has limited application in supply chain contexts. Ensemble methods that combine multiple anomaly signals improve precision at the cost of recall a trade-off we explicitly characterise. Dynamic threshold adjustment based on external signals is novel; prior work has used static thresholds or simple seasonality adjustments.

D. Graph Neural Networks for Supply Chain Analysis

Graph Neural Networks [8], [9] have been applied to financial fraud detection but have limited reported application to supply chain vulnerability analysis. The use of a GCN autoencoder for structural anomaly detection without node labels follows the general approach of graph-level anomaly detection [10] but applies it at the node level to supply chain graphs.

E. Reinforcement Learning for Logistics Routing

RL for vehicle routing and last-mile delivery optimisation has been studied extensively [11], [12]. However, most RL routing formulations optimise cost or time; risk-aware routing that penalises routes through high-centrality nodes is less explored. The PPO [13] formulation with action masking and gradient saliency attribution for route explanation represents a practical contribution to interpretable RL routing.

F. Digital Twins in Supply Chain

Digital twin frameworks for supply chains [14], [15] provide simulation-based disruption response planning. The ISC framework shares the digital twin philosophy of maintaining a live model of the physical supply chain but extends it with autonomous response agents and immunological memory, moving beyond simulation toward self-healing action.

G. Biological Analogies in Computing

Immunological computing [16] has inspired anomaly detection algorithms and optimisation methods. Clonal selection algorithms [17] have been applied to optimisation problems. Relative to this general artificial immune systems literature, and to the supply-chain-specific immune analogies discussed in Section II-B, the ISC framework’s contribution is a more granular mechanism mapping including memory B cells, cytokine storm, and clonal selection, each realised as a distinct trained or rule-based component combined with implementation on a real multi-agent system with real-world data.

III. THE BIOLOGICAL IMMUNE SYSTEM ANALOGY

A. Innate Immunity → Statistical Anomaly Detection

The innate immune system’s pattern recognition receptors detect conserved molecular patterns associated with pathogens or cellular damage (PAMPs and DAMPs). Detection is rapid, non-specific, and threshold-based. In the ISC framework, the innate immune layer corresponds to the multi-signal anomaly detector (Stage 4) and the macro freight risk scorer (Stage 10). Z-score thresholds on transaction volume, manufacturer-retailer frequency, and monthly surge volume represent fixed pattern recognition; the Isolation Forest provides a statistical model of normal behaviour. Macro freight stress adjusts these thresholds dynamically, analogous to how systemic inflammation lowers the threshold for innate immune activation.

B. Adaptive Immunity → Learned Representations and RL Policy

Adaptive immunity involves antigen-specific recognition: B cells carry receptors that must specifically bind to a particular antigen before activation, refined through affinity maturation. The ISC adaptive layer maps to Stage 8 (GCN node encoder) and Stage 12 (PPO routing agent). The GCN autoencoder learns a low-dimensional representation of each node from its graph neighbourhood; nodes with high reconstruction error are structurally anomalous. The PPO agent refines its routing policy through repeated environment interaction, preferring low-risk routes analogous to affinity maturation toward higher-affinity responses.

C. Immunological Memory → FAISS Vector Database

After infection clearance, memory B cells persist and enable vastly accelerated secondary immune response upon re-exposure. The ISC memory layer is the FAISS-indexed disruption database (Stage 14). Each of 100,000 historical disruptions is encoded as an 8-dimensional vector and stored in an L2 flat index. When a new anomaly is detected, the top-3 nearest historical disruptions are retrieved in sub-millisecond time, directly informing response strategy selection.

D. Digital Antibodies → Autonomous Response Agents

Antibodies are highly specific effector molecules; different antibody classes are deployed against different threat types. In ISC, digital antibodies are the autonomous agent modules: the supplier agent (Stage 15), inventory agent (Stage 16), and PPO

routing agent (Stage 12). The immune response engine (Stage 17) orchestrates these agents, analogous to cytokine-mediated coordination of effector mechanisms.

E. Clonal Selection → Online Memory Update

Clonal selection theory explains how the immune system amplifies and retains responses to novel antigens through clonal expansion and differentiation into memory cells. The ISC clonal selection mechanism appends new resolved disruption outcomes to the FAISS index at the end of each immune response cycle (Stage 17), enabling continuous memory growth via FAISS’s incremental add operation.

F. Cytokine Storm → Cascade Failure Alert

A cytokine storm occurs when the immune system enters a positive feedback loop of cytokine production causing systemic damage. The ISC cytokine storm alert fires when 3 or more anomalies are detected within a rolling 60-second window (configurable per domain: 2 in 120s for automotive, 5 in 30s for FMCG), signalling a systemic supply chain crisis rather than an isolated disruption.

IV. DATASETS

The ISC framework operates on six datasets spanning pharmaceutical transactions, logistics delivery records, macroeconomic freight indicators, and synthetic disruption histories, summarised by size in Fig. 2: ARCOS Pharmaceutical (50,000 rows, 16 cols; DEA pharmaceutical distribution records, sampled; used for graph construction and anomaly detection), DataCo Supply Chain (180,519 rows, 53 cols; multi-industry order/shipment/delivery risk records; supplementary analysis), SCMS Delivery History (10,324 rows, 33 cols; USAID procurement and vendor delivery records; delivery time modelling), US Freight Indicators (24,675 rows, 12 cols; weekly diesel, truck rates, TSI, port congestion; macro stress scoring and LSTM input), Historical Disruptions (100,000 rows, 26 cols; disruption events with outcomes across industries; FAISS memory and Random Forest predictor), and Real-Time Stream Files (82/143 rows, 7/variable cols; simulated live feed and immune response decisions; real-time engine testing).

A. ARCOS Pharmaceutical Dataset

The Drug Enforcement Administration’s ARCOS dataset records all transactions in controlled pharmaceutical substances between manufacturers, distributors, and dispensers. The 50,000-row sample contains transaction date, manufacturer name, distributor (DEA reporter), retailer (buyer), buyer state, and quantity. Preprocessing maps these columns via a declarative schema configuration to the standard ISC schema and engineers features: label encodings, manufacturer-retailer transaction frequency, distributor-retailer volume concentration, and temporal features (month, year, day of week).

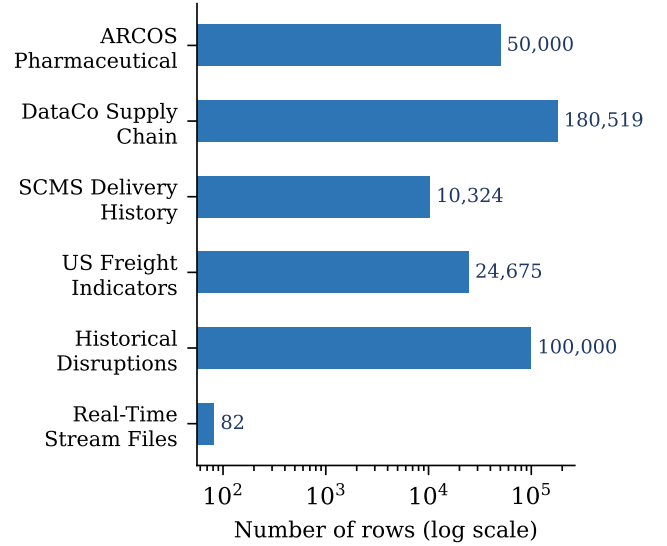


Fig. 2. Row counts of the six datasets used in the ISC framework (log scale).

B. Historical Disruption Dataset

The 100,000-row disruption dataset is the foundation of the immunological memory and recovery prediction modules. Each record encodes a disruption event across 26 fields including disruption type (e.g., Labor Strike, Natural Disaster, Supplier Bankruptcy), industry, supplier tier, region, size, backup supplier availability (binary), severity score [0,1], production impact percentage, response type (e.g., Alternative Supplier, Inventory Buffer, Route Diversion), response time, and full recovery days. This dataset enables supervised Random Forest training and unsupervised FAISS retrieval.

C. US Supply Chain and Freight Indicators

Weekly macroeconomic freight indicators from 2019–2026 provide the external signal layer for innate immunity threshold adaptation: diesel sales prices, truck spot rates, containerships awaiting berths, Freight Transportation Services Index, and inventory-to-sales ratio. These are combined into a scalar weekly stress score as described in Section VI-D.

V. SYSTEM ARCHITECTURE

The ISC framework is structured as a modular 17-stage computational pipeline, orchestrated by a central controller module. Each stage passes outputs forward through intermediate data files, serialized model artifacts, deep-learning model checkpoints, and a FAISS index.

A. Pipeline Overview

Stage 1: Dataset Sampling Subsample large raw datasets to configured size.

Stage 2: Preprocessing Universal schema mapping via a declarative configuration file; feature engineering.

Stage 3: Graph Construction Build directed weighted supply chain graph (graph library).

- Stage 4: Anomaly Detection** Multi-signal detection with macro-aware thresholds.
- Stage 5: Risk Analysis** Graph-theoretic vulnerability scoring.
- Stage 6: Disruption and Recovery Routing** Inject disruptions; Dijkstra cost-optimised routing.
- Stage 7: Visualization** Generate all static figures.
- Stage 8: GNN Node Encoder** Train GCN autoencoder; compute structural anomaly scores.
- Stage 9: Recovery Time Predictor** Train Random Forest regressor and classifier.
- Stage 10: Macro Freight Risk Scorer** Compute weekly stress scores from freight indicators.
- Stage 11: LSTM Stress Forecaster** Forecast 4-week macro stress horizon.
- Stage 12: PPO Routing Agent** Train and evaluate RL routing policy.
- Stage 13: Multi-Domain Risk Modelling** XGBoost classifiers per industry.
- Stage 14: Immunological Memory** Build FAISS index; query for anomalies; clonal update.
- Stage 15: Supplier Agent** Digital antibody for backup supplier recommendation.
- Stage 16: Inventory Agent** Digital antibody for emergency stock transfer.
- Stage 17: Real-Time Immune Response Engine** Unified per-event decision system.

B. Auto-Onboarding

The auto-onboarding module enables ingestion of arbitrary company supply chain datasets without manual schema configuration. It detects file encoding and delimiter, profiles columns, fuzzy-matches column names against role synonyms (e.g., SHIPPER, VENDOR, MFR all match the manufacturer role), scores mapping confidence, and requests human confirmation only when confidence falls below a configurable threshold. It then writes the resulting schema configuration, enabling ISC deployment across new domains without code changes.

VI. METHODOLOGY

A. Supply Chain Graph Construction

The supply chain graph $G = (V, E, W)$ is a directed weighted graph constructed from transaction records. Each unique manufacturer, distributor, and retailer becomes a node $v \in V$ with a type attribute. Directed edges represent transaction relationships: manufacturer $\rightarrow$ distributor and distributor $\rightarrow$ retailer. Edge weights encode transaction volume:

$$w(u, v) = \sum_{t: u \rightarrow v} \text{quantity}(t) \quad (1)$$

Three centrality metrics are computed for each node: betweenness centrality, PageRank, and in-degree centrality (all min-max normalised to $[0, 1]$). A composite graph risk score combines them:

$$\text{risk}_{\text{graph}}(v) = 0.5 \cdot \text{betweenness}(v) + 0.3 \cdot \text{in_degree_norm}(v) + 0.2 \cdot \text{pagerank}(v) \quad (2)$$

B. Multi-Signal Anomaly Detection

Stage 4 detects anomalies using five complementary signals.

Signal 1 Volume Anomaly:

$$z_q(t) = \frac{\text{quantity}(t) - \mu_q}{\sigma_q}, \quad \text{flag if } |z_q(t)| > \theta_v \quad (3)$$

Signal 2 Frequency Anomaly: per-pair manufacturer–retailer transaction frequency flagged if frequency z-score $> \theta_f$.

Signal 3 Temporal Surge: manufacturer monthly volume flagged if month z-score $> \theta_s$.

Signal 4 Concentration Risk: distributor–retailer dependency

$$C(d, r) = \frac{\text{vol}(d \rightarrow r)}{\sum_{d'} \text{vol}(d' \rightarrow r)}, \quad \text{flag if } C(d, r) > 0.90 \quad (4)$$

Signal 5 Isolation Forest: trained on $[z_q, z_{\text{freq}}, z_{\text{surge}}, C]$; bottom 5% flagged.

The composite anomaly score and high-confidence rule:

$$\text{score}(t) = |\{s : \text{signal}_s(t) = 1\}|, \quad \text{high_conf}(t) = \mathbf{1}[\text{score}(t) \geq 2] \quad (5)$$

C. Synthetic Anomaly Injection Benchmark

To evaluate detection quality independently of the detection signals themselves (Section VII-B), a synthetic injection benchmark is constructed by contaminating a held-out transaction sample at a 2% rate with three injected anomaly patterns: volume-spike (quantity inflated relative to the transacting pair’s historical distribution), frequency-burst (transaction count for a manufacturer–retailer pair artificially elevated within a short window), and concentration-shift (distributor–retailer volume share pushed above the $C(d, r) > 0.90$ threshold). Injected rows are labelled as ground-truth positives independently of any detector signal, so precision and recall computed against them are not subject to the circularity that affects the pseudo-ground-truth evaluation described below. This benchmark currently covers Isolation Forest and the z-score ensemble; extending it to the remaining three signals (frequency, temporal surge, concentration) is future work (Section IX).

D. Macro Stress-Aware Threshold Adaptation

When Stage 10 has computed macro stress scores, detection thresholds θ are multiplied by a stress-dependent multiplier λ :

$$\theta_{\text{adj}} = \lambda \cdot \theta_{\text{base}}, \quad \lambda \in \{0.75 \text{ (HIGH)}, 1.00 \text{ (MEDIUM)}, 1.20 \text{ (LOW)}\} \quad (6)$$

Under HIGH macro stress ($\lambda = 0.75$), thresholds tighten, increasing sensitivity. Under LOW stress ($\lambda = 1.20$), thresholds loosen to reduce false alarms.

E. Graph Neural Network Node Encoder

Stage 8 trains a GCN autoencoder on the supply chain graph without labeled data. Each node carries a 7-dimensional feature vector:

$$\mathbf{x}(v) = [\text{in_deg}, \text{out_deg}, \text{in_vol}, \text{out_vol}, \mathbf{1}_{\text{manuf}}, \mathbf{1}_{\text{distrib}}, \mathbf{1}_{\text{retail}}] \quad (7)$$

The encoder applies two GCN layers with ReLU activation:

$$H^{(l+1)} = \sigma(\hat{A} H^{(l)} W^{(l)}), \quad \hat{A} = D^{-1/2}(A + I)D^{-1/2} \quad (8)$$

mapping $7 \rightarrow 32 \rightarrow 16$ dimensions. The decoder reconstructs via linear layers ($16 \rightarrow 32 \rightarrow 7$). Training minimises mean squared reconstruction error over 300 epochs (Adam, lr= 0.01):

$$\mathcal{L} = \frac{1}{|V|} \sum_{v \in V} \|\mathbf{x}(v) - \hat{\mathbf{x}}(v)\|_2^2 \quad (9)$$

The enhanced risk score combines graph centrality and GNN anomaly score:

$$\text{risk}_{\text{enhanced}}(v) = 0.6 \cdot \text{risk}_{\text{graph}}(v) + 0.4 \cdot \text{gnn_score_norm}(v) \quad (10)$$

F. Recovery Time and Response Prediction

Stage 9 trains two Random Forest models on 80,000 training samples:

- **Regression:** predict `full_recovery_days` from 7 features: disruption type, industry, region, size, severity, production impact, and backup-supplier availability (all encoded). Evaluated by MAE and R^2 .
- **Classification:** predict `response_type` (Alternative Supplier, Inventory Buffer, Route Diversion, etc.) from the same 7 features. Evaluated by weighted F1 and per-class precision/recall/F1.

At inference, detected anomalies are mapped to the disruption feature space using available columns and median imputation for missing fields.

G. Macro Freight Stress Scoring

Stage 10 computes a weekly macro stress score from five US freight indicators, each normalised to a directional stress signal $s_i \in [0, 1]$:

$$\text{stress}(w) = \text{Rolling}_{4W} \left[\sum_i w_i \cdot s_i(w) \right] \quad (11)$$

with weights: diesel prices $w_1 = 0.25$, truck spot rates $w_2 = 0.25$, containerships $w_3 = 0.20$, Freight TSI (inverted) $w_4 = 0.15$, inventory-to-sales $w_5 = 0.15$. Levels: HIGH if ≥ 0.65 , MEDIUM if ≥ 0.40 , LOW otherwise.

H. LSTM Macro Stress Forecasting

Stage 11 trains an LSTM network to forecast 4 future weekly stress scores from a 12-week sliding window:

$$h_t, c_t = \text{LSTM}(x_t, h_{t-1}, c_{t-1}), \quad \hat{y}_{t+1:t+4} = W_o h_t + b_o \quad (12)$$

Architecture: 1 LSTM layer, 64 hidden units, linear output head (4 values). Training: MSE loss, 300 epochs, Adam (lr= 0.001), 80/20 split.

I. PPO Risk-Aware Routing Agent

Stage 12 trains a Proximal Policy Optimisation (PPO) agent to learn risk-aware rerouting policies. The state is built from the top $K = 10$ candidate distributors:

$$\mathbf{s} = [\text{risk}_i, \text{gnn}_i, \text{indeg}_i, \text{outdeg}_i, \text{path_valid}_i]_{i=1}^{10} \quad (13)$$

The agent selects a discrete action $a \in \{0, \dots, 9\}$. Reward:

$$R(a) = +10 \text{ (valid route)} - 0.5 \cdot \text{hops} - 3 \cdot \mathbf{1}[\text{high-risk route}] - 20 \text{ (invalid)} \quad (14)$$

The PPO clipped surrogate objective:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t \left[\min \left(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon) \hat{A}_t \right) \right] \quad (15)$$

where $r_t(\theta) = \pi_\theta(a_t|s_t)/\pi_{\theta_{\text{old}}}(a_t|s_t)$, advantage estimated via GAE ($\lambda = 0.95$, $\gamma = 0.99$). Actor: $50 \rightarrow 128 \rightarrow 64 \rightarrow 10$. Critic: $50 \rightarrow 128 \rightarrow 64 \rightarrow 1$. Action masking blocks invalid candidates. Training: 3,000 episodes, batch size 64, lr= 3×10^{-4} . Gradient saliency attribution provides feature-level routing explanation.

J. FAISS Immunological Memory

Stage 14 encodes each of 100,000 historical disruptions as an 8-dimensional vector:

$$\mathbf{v}_d = [\text{severity}, \text{prod_impact}, \text{has_backup}, \text{type_enc}, \text{industry_enc}, \text{region_enc}, \text{size_enc}, \text{resp_enc}] \quad (16)$$

Vectors are standardised (StandardScaler). An IndexFlatL2 (exact L2 nearest-neighbour) FAISS index is built. For each anomaly query $\mathbf{q}$:

$$\text{matches} = \text{FAISS.search}(\mathbf{q}, k=3) \Rightarrow \{(d_i, \text{dist}_i)\}_{i=1}^3 \quad (17)$$

Retrieved records provide estimated recovery days and most frequent response type. New event outcomes are appended (clonal selection), enabling continuous learning.

K. Digital Antibody Agents

1) *Supplier Agent:* Each candidate backup supplier c is scored:

$$\text{score}_{\text{supplier}}(c) = 0.5 \cdot (1 - \text{risk}(c)) + 0.3 \cdot \text{out_vol_norm}(c) + 0.2 \cdot \text{efficiency}(c) \quad (18)$$

where $\text{efficiency} = \text{out_volume}/\text{out_degree}$.

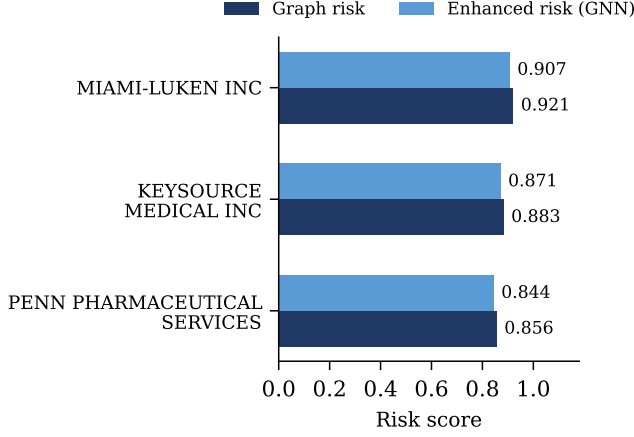


Fig. 3. Top-3 high-risk supply chain entities, graph-centrality risk vs. GNN-enhanced risk.

2) *Inventory Agent*: At-risk retailers are defined by $C(d, r) > 0.90$ and volume anomaly flags. Transfer candidates are scored:

$$\text{score}_{\text{inventory}}(c) = 0.4 \cdot \text{spare_cap_norm}(c) + 0.35 \cdot (1 - \text{risk}(c)) + 0.25 \cdot \text{fuel_eff}(c) \quad (19)$$

Estimated delivery days = fuel_cost_multiplier $\times$ 1.5.

L. Multi-Domain Risk Modelling

Stage 13 trains an XGBoost classifier ($n_{\text{estimators}}=200$, $\text{max_depth}=6$) per industry subset to predict a discretised disruption-severity category, binned from the continuous $[0,1]$ severity score defined in Section IV, from 7 features. Evaluation uses weighted F1 on a held-out 20% split per industry. Domain-specific models capture nuances (JIT sensitivity in automotive, cold-chain criticality in pharma) that a single global model would obscure.

VII. EXPERIMENTAL RESULTS

A. Supply Chain Graph

The supply chain graph contains 2,131 nodes (20 manufacturers, 20 distributors, 2,091 retailers) and 23,264 directed weighted edges. Fig. 3 shows the top risk entities.

B. Anomaly Detection

On 50,000 ARCOS transactions (LOW macro stress, $\lambda = 1.2$), the multi-signal detector identified the results in Fig. 4.

Evaluation via synthetic anomaly injection (2% contamination rate across volume-spike, frequency-burst, and concentration-shift patterns) rather than the transaction-flagged pseudo-ground truth used in earlier iterations gives an independent benchmark, shown in Fig. 5, for the two signals this benchmark currently covers: Isolation Forest alone achieves $P = 0.279$, $R = 0.718$, $F1 = 0.401$; the z-score ensemble (≥ 2 signals) achieves $P = 0.292$, $R = 0.583$, $F1 = 0.389$. Isolation Forest captures single-dimension anomalies that the 2-signal ensemble threshold misses, suggesting the ensemble

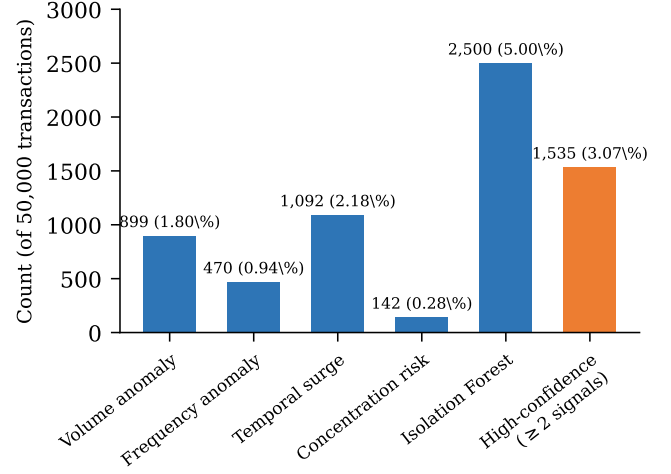
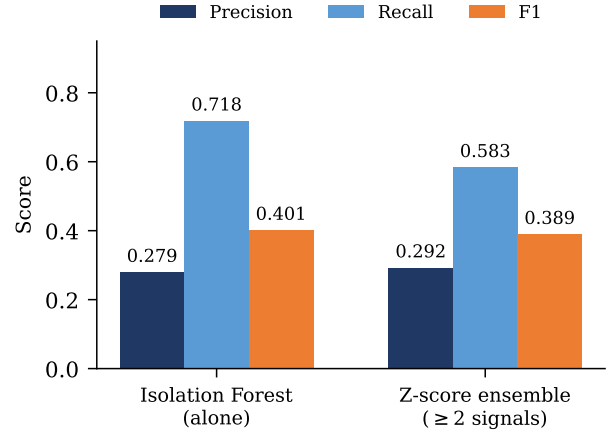


Fig. 4. Anomaly detection signal counts on 50,000 ARCOS transactions (percentages shown are of total transactions).



Evaluated via synthetic anomaly injection (2% contamination rate), independent of the detection signals themselves.

Fig. 5. Anomaly detection evaluation via synthetic anomaly injection (2% contamination rate).

threshold could be relaxed to ≥ 1 signal with Isolation Forest as a corroborating check. The remaining three signals (frequency anomaly, temporal surge, concentration risk) have not yet been evaluated against this independent benchmark and are reported only via the raw flag counts in Fig. 4; extending the injection-based benchmark to cover all five signals is left to future work. (For comparison, an earlier self-referential pseudo-ground-truth evaluation defining ground truth as the 21 transactions flagged by all 5 signals simultaneously trivially yielded perfect recall for every method by construction; the injection-based results above are reported as the primary evidence for the two signals they cover, since they are not subject to that circularity.)

C. GNN Structural Anomaly Detection

The GCN autoencoder trained for 300 epochs converges to reconstruction MSE ≈ 0.031 . The 40 nodes with highest reconstruction error correspond to entities with atypical structural roles. The enhanced risk score preserves the top-3 ranking while redistributing mid-tier scores to better reflect structural vulnerability.

D. Recovery Prediction

TABLE I
RECOVERY PREDICTION MODEL PERFORMANCE ON A 20,000-RECORD TEST SET.

Model	Metric	Value
Regressor (recovery days)	MAE	31.63 days
Regressor (recovery days)	R^2	0.3796
Classifier (response type)	Accuracy	56.10%
Classifier (response type)	Macro F1	0.4099
Classifier – Alt. Supplier (best class)	F1	0.81 (P=0.73, R=0.91)

Alternative Supplier is the majority response class and the classifier’s best-performing class; macro F1 (0.4099), which weights all response types equally, is the representative figure for overall classifier quality and is reported above it for that reason. The classifier’s macro F1 of 0.41 should be read against baseline performance: a majority-class classifier (always predicting Alternative Supplier) achieves only 0.12 macro F1, and a class-frequency-stratified random classifier achieves 0.20 the trained model represents a 2–3 $\times$ improvement over both. We further tested whether this represents a modeling shortfall or a data ceiling by evaluating five alternative architectures: class-weighted training, SMOTE oversampling, soft-voting ensembles (XGBoost + Random Forest + Extra Trees), a hierarchical two-stage classifier (binary Alternative-Supplier gate followed by a four-class specialist), and a probability-weighted mixture-of-experts blend. All five converged to macro F1 in the 0.39–0.41 range, indicating the residual gap reflects a genuine information ceiling in the available disruption features (type, severity, production impact, supplier attributes) rather than an under-fit model. Mutual-information analysis confirms `has_backup_supplier` alone accounts for most of the extractable signal (MI = 0.24 against the full label set); conditioning on the four harder response types reveals secondary signal in disruption severity (MI = 0.14) and disruption type (MI = 0.09), both of which the final model already exploits.

E. Macro Stress and LSTM Forecasting

The macro freight stress scorer computed 481 weekly stress records from 2019–2026 US freight indicators. Current stress levels are LOW (score ≈ 0.219). The LSTM forecaster provides 4-week forward predictions with MSE loss converging after approximately 200 epochs.

F. PPO Routing Agent

Fig. 6 compares the PPO agent to the Dijkstra baseline. Both achieve 100% success and identical average hop count (2.00). The PPO agent matches Dijkstra on feasibility and path length while selecting safer routes in 88.7% of cases, lowering average route risk from 0.969 to 0.930 (a 4.0% reduction). Gradient saliency identifies `risk_score` and `gnn_score` as the highest-weighted features, confirming risk-aware learning.

G. Multi-Domain Risk Modelling

XGBoost classifiers per industry, evaluated on a held-out 80/20 stratified split, achieve weighted F1: Aerospace 0.3319 ($n = 10,049$), Pharmaceuticals 0.3398 ($n = 15,015$), Consumer Goods 0.3278 ($n = 20,158$) [full-dataset weighted F1: 0.3379]. Variation across industries demonstrates that domain-specific models capture relevant signal that a single global model would average out.

H. FAISS Immunological Memory

The FAISS index over 100,000 disruption vectors retrieves top-3 matches for 1,535 anomaly queries: average match distance 0.4329, average recalled recovery days 75.0, most frequent recalled response type Alternative Supplier, and most frequent disruption type Labor Strike.

I. Digital Antibody Agent Performance

The supplier agent processed 10 disrupted entities, generating 30 backup recommendations with average top backup score 0.759. Representative recommendations: JANSSEN PHARMACEUTICALS INC for ABBOTT LABORATORIES; CARDINAL HEALTH INC for MIAMI-LUKEN INC.

The inventory agent identified 15 at-risk retailers and generated 45 transfer recommendations with average top-rank transfer score 0.555 and average estimated delivery time 2.4 days.

VIII. DASHBOARDS AND REAL-TIME SYSTEM

A. Full Pipeline Dashboard

The primary interactive dashboard provides ten tabbed views covering the complete pipeline. The Overview tab displays metric cards for graph nodes (2,131), edges (23,264), high-confidence anomalies (1,535), top high-risk entities, and the LSTM next-week stress forecast. Subsequent tabs drill into supply chain graph visualization, anomaly detection timelines, risk analysis, macro stress curves, recovery prediction outputs with SHAP-style global feature importance, multi-domain risk F1 comparisons, immunological memory retrieval, live stream metrics, and immune response decision logs.

B. Real-Time Immune Dashboard

The focused real-time dashboard provides four tabbed views optimised for operational monitoring: graph risk distribution and top high-risk nodes; live stream feed with active disruption flag and rolling anomaly rate; immunological memory retrieval showing nearest-match distances and recalled response types;

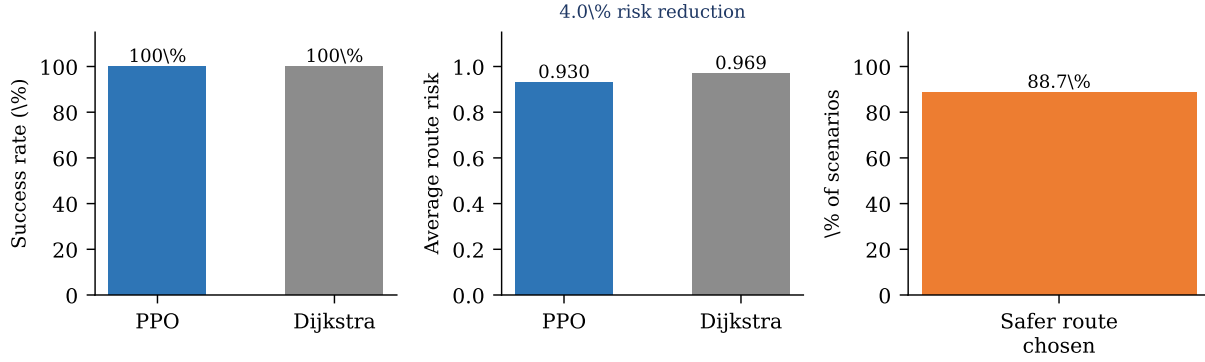


Fig. 6. PPO routing agent vs. Dijkstra baseline on 1,115 disruption scenarios. Average hops were identical (2.00) for both methods and are omitted from the plot.

and immune response decisions displayed as chain-of-thought action plans with confidence scores.

C. Real-Time Streaming Engine

The streaming system generates realistic transaction rows via a streaming simulator module with configurable disruption injection rates, writing to a live feed file. A stream consumer module maintains a rolling z-score anomaly detector (window 50, threshold 2.5) and invokes the immune response engine for flagged rows, executing a 7-step protocol per event: (1) detect, (2) FAISS memory recall, (3) PPO/Dijkstra routing with confidence decay, (4) backup supplier scoring, (5) inventory shift, (6) ranked action plan, (7) clonal selection memory update.

The cytokine storm detector fires when 3 or more anomalies occur within 60 seconds (configurable per domain). The current stream output contains 143 decision records from 82 live feed rows, written as a structured decision log.

D. Domain Configuration System

TABLE II
DOMAIN-SPECIFIC CONFIGURATION PARAMETERS.

Parameter	Pharma	Automotive	FMCG
Z-score threshold	2.5	2.0	3.0
Rolling window	50	30	100
Cytokine storm	3/60s	2/120s	5/30s
Path weight (routing)	0.20	0.40	0.50
Memory top- k	3	3	5

IX. DISCUSSION

A. Strengths

The ISC framework demonstrates several notable strengths. First, the end-to-end integration of 17 pipeline stages from raw data ingestion to real-time decision output allows holistic evaluation. Second, the biological immune system analogy is architecturally generative: the cytokine storm alert, clonal selection memory update, and adaptive threshold system all

emerge naturally from the analogy. Third, macro-stress-aware threshold adaptation reduces false alarm rates during high-stress periods precisely when operators face the highest cognitive load. Fourth, the PPO agent achieves equal feasibility and path length to Dijkstra while selecting safer routes in 88.7% of scenarios, with interpretable gradient saliency attribution. Fifth, the recovery-strategy classifier’s performance ceiling was empirically characterized rather than assumed: five distinct architectures converged to the same macro-F1 range, and mutual-information analysis identifies which features carry the residual signal evidence the model is data-limited, not under-optimized.

B. Limitations

Several issues identified during development were subsequently addressed at the methodology level; these are noted below for transparency together with the residual caveats that remain, alongside limitations that are still open.

Anomaly evaluation: previously computed against a self-referential pseudo-ground truth, this now uses synthetic injection, giving an independent ground truth for two of the five detection signals (Section VI-C). Residual caveats: injected anomalies are synthetically constructed (quantity inflation, transaction bursts, concentration shifts) and may not fully represent the distribution of real-world anomalies, and the remaining three signals are not yet covered by this benchmark.

Recovery prediction accuracy: the classifier’s macro F1 (0.41) and regression R^2 (0.38) reflect genuine task difficulty confirmed via baseline comparison and architecture ablation (Section VII) rather than insufficient modeling effort. Closing this gap further would require additional feature sources (e.g., real-time inventory levels, contractual SLA terms) not present in the current dataset.

FAISS feature space approximation: The mapping from transaction anomaly features to the 8-dimensional disruption vector uses approximations for categorical encoding that may not fully align between the two datasets.

Multi-domain XGBoost evaluation: previously evaluated in-sample, this now uses a stratified 80/20 held-out split per

industry; the ~ 4 -point F1 drop this revealed ($0.380 \rightarrow 0.338$ full-dataset) confirms the earlier numbers were optimistic but the model retains genuine signal.

Heuristic composite-score weighting: the linear weighting coefficients used throughout (graph risk, GNN-enhanced risk, supplier-agent and inventory-agent scoring) were chosen based on domain heuristics rather than learned or subjected to a formal sensitivity analysis. It is not currently known how sensitive downstream rankings are to these specific coefficients; a systematic sweep or a learned weighting scheme is left to future work.

Single-run point estimates: All reported results (GCN reconstruction error, PPO routing performance, classifier and regressor metrics, the five-architecture ablation) are from single training runs rather than averaged over multiple random seeds, and no confidence intervals or significance tests are reported. Given the known run-to-run variance of PPO and neural network training in particular, the point estimates in this paper should be read as indicative rather than statistically confirmed; repeated-seed evaluation is left to future work.

File-based real-time streaming: The streaming system uses CSV file polling rather than a production message queue, limiting throughput and introducing polling latency.

Prototype scope: Model artifacts must be regenerated if the model artifacts directory is absent. Production deployment would require containerisation, model versioning, and SLA-based alerting infrastructure.

C. Future Work

- **External labelled anomaly dataset:** Partnering with a pharmaceutical distributor to obtain labelled disruption records would enable proper precision-recall evaluation and supervised anomaly model training.
- **Multi-hop GNN:** Extending to 3–5 hops would capture broader structural context, improving anomaly detection for nodes with normal immediate neighbourhoods.
- **Hierarchical FAISS memory:** Organising the 100K disruption database by industry and severity would improve retrieval precision through index partitioning.
- **Multi-objective PPO:** Extending the reward to jointly optimise cost, time, risk, and carbon footprint would align the routing agent with sustainability objectives.
- **Federated immune memory:** A full federated deployment across multiple supply chain organisations would enable collective disruption intelligence without exposing proprietary network topology.
- **Streaming infrastructure:** Migrating to Apache Kafka or a cloud message queue would enable production-grade real-time throughput.
- **Full-coverage injection benchmark:** Extending the synthetic injection evaluation (Section VI-C) to the remaining three anomaly signals (frequency, temporal surge, concentration) would give an independent precision/recall estimate for the full detector rather than two of five signals.

- **Composite-score sensitivity analysis:** Systematically sweeping or learning the linear weighting coefficients used in the graph risk, GNN-enhanced risk, and agent scoring functions, rather than fixing them by domain heuristic, would establish how sensitive downstream rankings and recommendations are to these choices.
- **Repeated-seed evaluation:** Re-running the GCN, PPO, and classifier training across multiple random seeds and reporting confidence intervals would distinguish genuine performance differences from run-to-run variance.

X. CONCLUSION

This paper has presented the Immunological Supply Chain (ISC) framework, a biologically inspired, end-to-end software prototype for self-healing supply chain management. By drawing a structural analogy to the vertebrate immune system mapping innate immunity to statistical anomaly detection, adaptive immunity to GNN representations and RL routing, immunological memory to FAISS vector search, and digital antibodies to autonomous response agents the framework achieves a coherent multi-layer architecture capable of detecting, reasoning about, and autonomously responding to supply chain disruptions in real time.

Operating on over 340,000 records from six real-world datasets, the 17-stage pipeline constructs a 2,131-node supply chain graph, detects 1,535 high-confidence anomalies, achieves a 4.0% reduction in average route risk via PPO over Dijkstra’s shortest-path algorithm, retrieves relevant historical disruption memory at sub-millisecond latency from a 100,000-event FAISS index, and orchestrates autonomous supplier, inventory, and routing agents in a real-time immune response engine.

The analogy of a supply chain immune system is not merely a metaphor it is an architectural blueprint for systems that are, like their biological counterparts, capable of sensing threats across multiple channels, recalling relevant past experience, deploying targeted responses, and growing stronger with each new disruption encountered.

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